

Cursed PCB

Four addressable RGB LEDs, onboard USB-C power, and built-in SerialUPDI programming. This card gets the example running; the full guide and code live on GitHub.



Before you start

- Use a USB-C **data** cable, not a charging-only cable.
- Install Arduino IDE from arduino.cc/en/software.
- No separate LED library is needed; megaTinyCore includes it.
- For external power, use side-header **VCC + GND**—never feed the 5V pin.

Install the ATtiny board package

- 1 In Arduino IDE, open **File** → **Preferences** (Arduino IDE → Settings on macOS).
- 2 Add this to **Additional Boards Manager URLs**:

```
http://drazzy.com/package_drazzy.com_index.json
```

- 3 Open **Tools** → **Board** → **Boards Manager**, search for **megaTinyCore**, and install the newest version by Spence Konde.

Connect and choose these settings

Plug the board into the computer. The power LED should turn on and a new serial port should appear.

Tools menu	Choose
Board	ATtiny412/402/212/202 (without Optiboot)
Clock	20 MHz internal
BOD Voltage Level	4.2V (20 MHz or less)
Programmer	SerialUPDI - 230400 baud

Make some light

Get and upload the example

Code and full instructions

github.com/user-will/cursedPCB

- 1 Download the repository and open `examples/LED_example/LED_example.ino`.
- 2 Choose **Sketch** → **Verify/Compile**.
- 3 Choose **Sketch** → **Upload Using Programmer**. Do not use the regular Upload command.

The LEDs show red, green, blue, white, a rainbow sweep, and a moving white chase.

Optional mode button

Connect a normally-open pushbutton between header pins **0 (PA6)** and **2 (PA1)**. No external resistor is needed with the example sketch. Each press advances the animation. The LEDs still work without the button.

Six-pin header

Component side up, USB-C at the bottom; read left to right:

Label	ATTiny	Use
5V	VDD	Regulated 5 V output; not a power input
GND	GND	Ground
0	PA6	Digital, DAC, TX
1	PA7	Digital, RX
2	PA1	Digital, I²C SDA
3	PA2	Digital, I²C SCL

The LEDs use **PA3 / Arduino pin 4**. PA0 is reserved for UPDI programming. The side three-pin header is **UPDI, GND, VCC** from top to bottom; VCC is the external-power input. The onboard USB connection programs UPDI; it is not a Serial Monitor connection.

Troubleshooting

- **No port:** try a known data cable or install a CH340 driver.
- **Upload error:** recheck board, BOD voltage, and programmer; close Serial Monitor.
- **Still failing:** select *SerialUPDI - SLOW: 57600 baud*.
- **Dark LEDs:** use PIN_PA3, four pixels, and NEO_GRB.

Power safely

Use USB-C normally. For standalone external power, connect positive to side-header **VCC** and negative to **GND**. Never power through the 5V pin or connect USB-C and VCC together. VCC feeds an L78L05: fixed 5 V, 100 mA total maximum, about 7 V minimum input for regulation, and 30 V absolute maximum. Higher input voltage creates more heat. High/full-white LED brightness can exceed 100 mA.